

Total No. of Questions :6]

SEAT No. :

P64

OCT. -16/BE/Insem. - 118

[Total No. of Pages :2

B.E. (Mechanical)

TRIBOLOGY

(2012 Course) (Semester -I) (Elective-I) (402044 B)

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Write Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 2) Figures to the right indicate full marks.*
- 3) Assume suitable data whenever necessary.*

Q1) a) Define Tribology. Mention minimum four tribological adverse effects generally arises in Industry. **[4]**

b) Compare sliding and rolling contact bearing in terms of the following. **[6]**

- i) Magnitude of load
- ii) Starting friction
- iii) Nature of the load
- iv) Positional accuracy
- v) Speed
- vi) Noise

OR

Q2) a) What is the effect of temperature and pressure on viscosity of lubricating oil? **[4]**

b) What is lubricant? Explain five desirable properties of lubricating oil used for bearings. **[6]**

Q3) What do you mean by Adhesive wear? Derive the Archard's equation for volume of Adhesive wear [Wear rate]. Write four assumptions in Archard's equation for Adhesive wear. **[10]**

OR

P.T.O.

- Q4)** a) Define friction and wear. Explain different laws of friction. [5]
b) Explain the stick-slip friction phenomenon. Write at least two examples. [5]

Q5) A short hydrodynamic journal bearing refers the following data: [10]

Journal speed = 35 revolutions per seconds (rps)

Length of bearing (l) = $0.5 \times$ Journal diameter (d)

Radial clearance (c) = $0.001 \times$ Journal diameter (d)

Eccentricity ratio (ϵ) = 0.65

Flow rate of Lubricant (Q_s) = 3.45 litre per hour

Radial Load (W) = 1000 N

Calculate:

- a) Journal Diameter
- b) Radial clearance
- c) Dimensions of the bearing
- d) Minimum oil film thickness
- e) Absolute viscosity of the lubricant

OR

- Q6)** a) With neat sketch, explain the mechanism of pressure development in hydrodynamic journal bearing. [6]
b) Write any four assumptions made while deriving Reynolds equation. Also write any four types of hydrodynamic thrust bearing. [4]

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